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Sex Differences in Sport and Exercise

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ACTIVITY 19.2 *Menstrual Dysfunction* describes the characteristics of delayed menarche, primary amenorrhea, secondary amenorrhea, and oligomenorrhea.

VIDEO 19.1 presents Jim Pivarnik on the benefits of exercise during pregnancy.

ACTIVITY 19.3 *Eating Disorders and Disordered Eating* explores the characteristics of anorexia nervosa and bulimia nervosa and the differences between eating disorders and disordered eating.

VIDEO 19.2 presents Nancy Williams on exercise and reproductive health and the causes of the female athlete triad.

In Closing

Amber Miller accomplished two amazing feats during the same weekend in October 2011. The 27-year-old marathoner ran the Chicago Marathon while nearly 39 weeks pregnant—and then gave birth to a baby girl hours later! Because of her training and running history, Miller received clearance from her doctor to participate, provided that she took a half-run, half-walk approach, drank plenty of fluids, and ate food along the course. She finished in just under 6.5 h. Miller had intended to stop halfway, but once she and her husband started running, they just kept going. After the race, she grabbed some food and headed to the hospital.

In 2014 and 2015, Olympian Alysia Montañó ran an 800 m race and runner Amy Keil competed in the Boston Marathon, both nearly 8 months into their pregnancies. Although these feats may seem extreme and should not be attempted by most pregnant women, physicians generally agree that if a woman is healthy and running regularly before she got pregnant, running during pregnancy is fine. Marathon world record holder Paula Radcliffe ran 14 mi (22.5 km) a day while pregnant and resumed training weeks after the birth of her first child. She won the New York City Marathon in 2007 just 10 months after giving birth.

It is unfathomable now that before 1972, girls and women were banned from official participation in most mainstream marathons, including the Boston Marathon!

In the past, it was not uncommon for young girls to be discouraged from participating in vigorous physical activity, while young boys climbed trees, raced against each other, and participated in various sports. The underlying notion was that boys were meant to be active and athletic but girls were less suited to physical activity and competition. Physical education classes furthered this idea by having girls perform nonstrenuous exercise activities. In sports, girls and women were not allowed to run in long-distance races, and basketball was limited to half-court, with each team having only offensive or defensive players.

With the advent of Title IX, athletic activities and programming must be equally accessible to girls and women, and the results have been amazing. Their athletic accomplishments parallel those of boys and men, with performance differences of 15%-17% or less for most sports and events. This is illustrated in [table 19.1](#), where men's and women's world records are compared for representative events in both track and field and swimming. Do these performance

differences represent true biological differences, or are there other factors involved? The focus of this chapter is on the extent to which biological differences between women and men affect performance capacity.

Sex Versus Gender in Exercise Physiology

The literature examining physiological differences between men and women refers to these comparisons as *sex differences* or *gender differences*, with the terms often used interchangeably. The dual terminology can cause confusion with regard to literature searches, data reporting, and interpretation of results, leading researchers to consider whether there is a distinction between the two terms and when each should be used.

TABLE 19.1 Selected Men's and Women's World Records Through 2018*

Event	Men	Women	Difference
Track and field			
100 m	9.58 s	10.49 s	9%
1,500 m	3:26.00 min:s	3:50.07 min:s	11%
10,000 m	26:17.53 min:s	29:17.45 min:s	11%
High jump	2.45 m	2.09 m	16%
Long jump	8.95 m	7.52 m	17%
Swimming (freestyle)			
100 m	46.91 s	51.71 s	10%
400 m	3:40.07 min:s	3:56.46 min:s	3%
1,500 m	14:31.02 min:s	15:20.48 min:s	5.5%

*Records current through June 2018, per iaaf.org and fina.org.

With respect to research on male and female physiology, sex and gender are not synonyms. Rather, sex is biologically determined and gender is culturally determined.²⁵ In other words, the term *sex* refers to the physiological, genetic, and biological traits that define a human as being male or female. Gender, by contrast, is the social, cultural, and psychological influences that construe an individual's self-representation as male or female. For example, it is possible for one to have the structural and functional characteristics of a man but to identify and live as a woman, and possibly to have undergone surgical transformation to associate with the female gender. Consequently, most basic physiological comparisons to date between men and women have established sex differences, unless

the investigation was specifically targeted at physiological outcomes associated with one or more of the cultural, social, behavioral, or psychological influences of different gender states. Although this seems to be a theoretical concept, real-life applications are observed in research and competitive sport.

For example, in 2011, the International Association of Athletics Federations (IAAF) devised a new policy in response to the case of Caster Semenya, a South African female runner who won the 800 m race at the 2009 Berlin World Championships. Semenya was criticized for her masculine build and forced to undergo sex testing that revealed unusually high levels of testosterone (a condition known as hyperandrogenism). The IAAF policy on hyperandrogenism thus states that women who are suspected of being too masculine must undergo testing and treatment in order to remain eligible for competition. This policy has been criticized on many points, among them that the development of the policy was confused and confounded by lack of distinction between sex and gender, as well as the difficulty of distinguishing easily between the two for the sake of competitive eligibility.¹⁶

Why the reluctance to use the term *sex differences* in modern-day physiology? It may be that the term *gender* avoids the innuendo associated with the term *sex*, and therefore seems less suggestive, more polite, or politically correct. In addition, even the concept of sex itself is not as dichotomous as one might wish, since genetic determinants of sex and biological structure and function can be discrepant so as to create a spectrum of male or female traits known as intersex characteristics in the same individual. In any case, though, referring to comparisons of the basic biological and physiological male and female attributes as sex differences establishes in the literature a basis for better classifying and interpreting research studies.

Body Size and Composition

Body size and composition are similar in boys and girls during early childhood. During late childhood, as we saw in [chapter 17](#), girls begin to accumulate more fat than boys. Starting in early

adolescence, boys begin to increase their fat-free mass (FFM) at a much higher rate than girls (see [figure 17.3](#)).

Body composition differences between the sexes occur primarily because of endocrine changes that begin during puberty. Before puberty, the anterior pituitary gland secretes very small amounts of the gonadotropic hormones: follicle-stimulating hormone (FSH) and luteinizing hormone (LH). During puberty, however, the anterior pituitary begins to secrete significantly greater amounts of both of these hormones. In girls, when sufficient quantities of FSH and LH are secreted, the ovaries develop and estrogen secretion begins. In boys, these same hormones trigger development of the testes and, in turn, testosterone secretion.

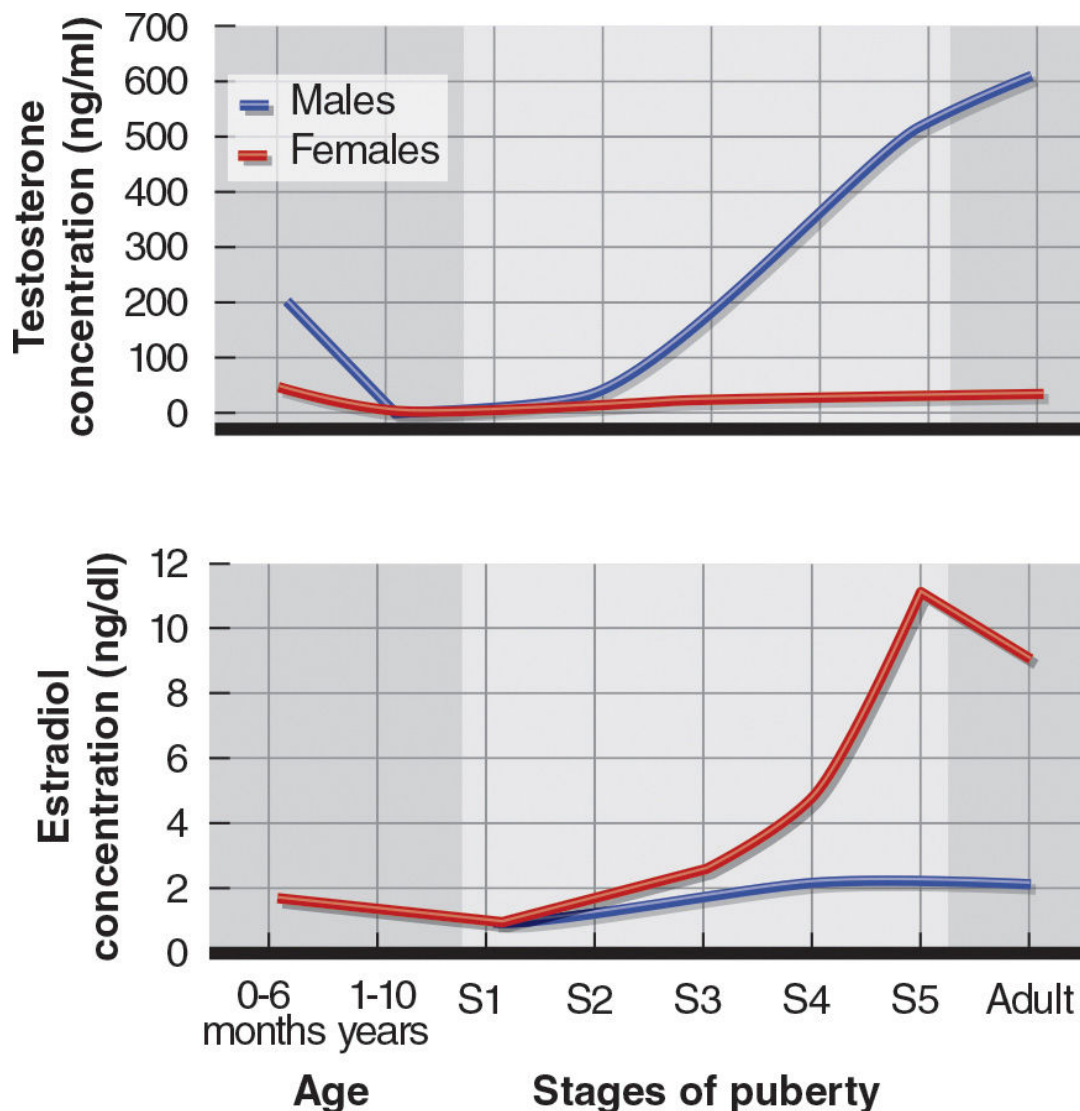


FIGURE 19.1 Changes in blood concentrations of testosterone and estrogen (estradiol) from birth to adulthood. S1 through S5 represent stages of puberty based on secondary sex characteristics, with S1 representing the initial stages of puberty and S5 the final stages.

Reprinted by permission from R.M. Malina, C. Bouchard, and O. Bar-Or, *Growth, Maturation, and Physical Activity*, 2nd ed. (Champaign, IL: Human Kinetics, 2004), 414; Data from Esoterix, Inc. *Endocrinology: Expected Values and S.I. Unit Conversion Tables*, 5th edition. (Calabasas Hills, CA: Esoteric, Inc., 2000).

Figure 19.1 illustrates these changes in estrogen (estradiol—the most potent form of estrogen) and testosterone from the beginning of puberty (S1) to the end of puberty (S5). Testosterone increases bone formation, which leads to larger bones, and increased protein synthesis, which in turn leads to increased muscle mass. As a result, adolescent boys are larger and more muscular than girls, and these characteristics continue into adulthood. At full maturity, not only do men have a greater muscle mass but also the distribution of the muscle mass differs from that of women. Men carry approximately 3% more muscle mass in the upper body than women. Testosterone also stimulates erythropoietin production by the kidneys, which leads to increased red blood cell production, as discussed later in this chapter.

Estrogen has a significant influence on women's body growth by broadening the pelvis, stimulating breast development, and increasing fat deposition, particularly in the thighs and hips. This increase in fat deposition in the thighs and hips is the result of increased **lipoprotein lipase** activity in these areas. This enzyme is considered the gatekeeper for storing fat in adipose tissue. Lipoprotein lipase is produced in the fat cells (adipocytes) but is bound to the walls of the capillaries, where it exerts its influence on the chylomicrons, which are the major transporters of triglycerides in the blood. When lipoprotein lipase activity in any area of the body is high, chylomicrons are trapped and their triglycerides are hydrolyzed and transported into the adipocytes in that area for storage.

Many women are constantly fighting fat deposition on the thighs and hips, but they are usually fighting a losing battle. Lipoprotein lipase activity is very high and lipolytic activity (fat breakdown) is low in the hips and thighs of women compared with women's other fat storage areas and compared with the hips and thighs of men. This results in a rapid storage of fat in women's thighs and hips, and the

decreased lipolytic activity makes it difficult for women to lose fat from these areas. During the last trimester of pregnancy and throughout lactation, the activity of lipoprotein lipase decreases and lipolytic activity increases dramatically, which suggests that fat is stored in the hips and thighs for reproductive purposes.



Estrogen also accelerates the growth rate of bone, allowing the final bone length to be reached within 2 to 4 years following the onset of puberty. As a result, girls grow very rapidly for the first few years following puberty and then cease to grow. Boys have a much longer growth phase, allowing them to attain a greater height.

Because of these differences, compared with fully mature men, fully mature women are on average about

- 13 cm (5 in.) shorter,
- 14 to 18 kg (30-40 lb) lighter in total weight,
- 3 to 6 kg (7-13 lb) heavier in fat mass, and
- 6% to 10% higher in relative body fat.

In Review

- The term **sex** refers to physiological, genetic, and biological traits. Gender, by contrast, is the social, cultural, and psychological influences that construe an individual's self-representation as male or female.
- Until puberty, girls and boys do not differ significantly in most measurements of body size and composition.
- At puberty, because of the influences of estrogen and testosterone, body composition begins to change markedly.
- Testosterone increases bone formation and protein synthesis, leading to a larger FFM. It also stimulates the production of erythropoietin, which increases red blood cell production.
- Estrogen causes increased fat deposition in women, particularly in the hips and thighs, and an increased rate of bone growth, such that bones in women reach their final length earlier than in men.

Physiological Responses to Acute Exercise

When women and men undergo an acute bout of exercise, whether an all-out run to exhaustion on the treadmill or a single attempt to lift the maximal weight possible, there are response differences between the sexes. Differences between children and adolescent boys and girls are discussed in [chapter 17](#). Here we focus on these differences in adults in the areas of strength and of cardiovascular, respiratory, and metabolic responses to exercise.

Strength

In terms of absolute strength, women are approximately 40% to 60% weaker than men in upper body strength but only 25% to 30% weaker in lower body strength. However, because of the considerable size difference between the average man and the

average woman, it is more appropriate to express strength either relative to body weight (absolute strength divided by body weight) or relative to FFM, as a reflection of muscle mass (absolute strength divided by FFM). When lower body strength is expressed relative to body weight, women are still 5% to 15% weaker than men, but when it is expressed relative to FFM, this difference disappears. This suggests that the innate qualities of muscle and mechanisms of neuromuscular control are similar for women and men, a fact that has been confirmed by computed tomography (CT) scans of the thighs and upper arms, that is, knee extensor muscles ([figure 19.2a](#)) and elbow flexor muscles ([figure 19.2b](#)).

Muscle biopsies have become more common among female athletes, allowing fiber type comparisons with male athletes in the same sport or event. From these data, men and women have similar fiber type distributions, although there is some evidence that men may reach greater extremes (greater than 80% type I or greater than 80% type II). Among elite distance runners,^{7,12} women had a mean value of 69% type I fibers compared with 79% for the men. The women predictably had much smaller type I and type II fibers (mean values of less than 4,500 μm^2 in women and greater than 8,000 μm^2 in men). Despite smaller fiber size in women, capillarization appears to be similar between men and women.

Research indicates that women have a greater resistance to fatigue compared with men. Fatigue is often tested by having subjects maintain a constant force at a given percentage of their maximal voluntary static action for as long as possible. As an example, women would be able to maintain a constant force output at 50% of their maximal static action for a longer period of time than men at the same 50% of their maximal static action. Men, being stronger, have to apply a greater absolute amount of force to achieve the same 50% relative force. The reason for this greater resistance to fatigue is not yet known but could be related to the amount of muscle mass recruited and the compression of blood vessels, substrate utilization, muscle fiber type, and neuromuscular activation.

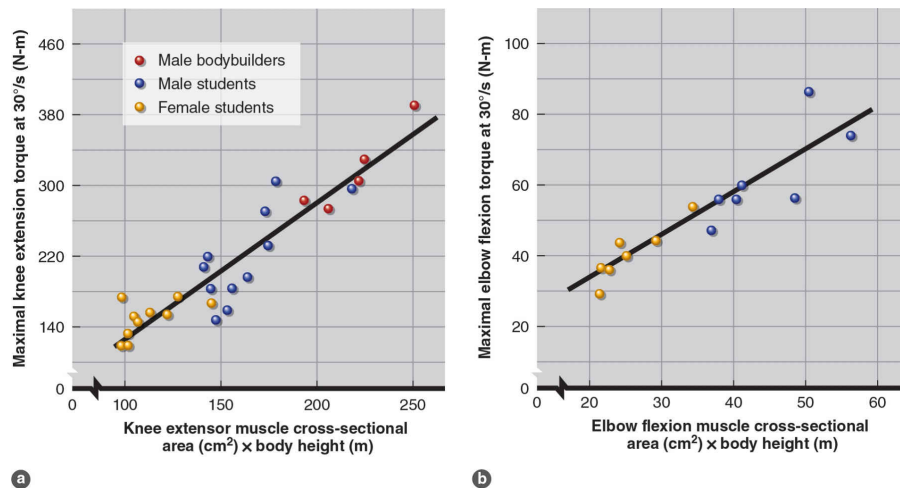


FIGURE 19.2 No sex differences between men and women in strength [(a) maximal knee extension torque or (b) maximal elbow flexion torque] are seen when strength is expressed per unit of muscle cross-sectional area.

Reprinted by permission from P. Schantz et al., "Muscle Fibre Type Distribution, Muscle Cross-Sectional Area and Maximal Voluntary Strength in Humans," *Acta Physiologica Scandinavica* 117 (1983): 219-226.



Cardiovascular and Respiratory Function

When exercising on a cycle ergometer at a fixed power output (independent of body weight), women generally have a higher heart rate (HR) response. Maximal heart rate (HR_{max}) is generally the same in both sexes. During submaximal exercise, stroke volume (SV) is lower in women, but HR is higher; therefore, cardiac output ($\dot{Q}$) is usually the same in men and women at any absolute submaximal power output. Remember that the *average* woman may also be less aerobically active and therefore less aerobically conditioned. However, even in fitness-matched men and women, the lower SV can be attributed to the following:

- Women have smaller hearts and therefore smaller left ventricles because of their smaller body size and possibly lower testosterone concentrations.

- Women have a smaller blood volume, which also is related to their size and lower FFM.

When power output is controlled to provide the same *relative* intensity of exercise, ($\% \dot{V}O_{2\max}$), women's heart rates are still slightly higher than men's, and their stroke volumes are markedly lower. At 60% $\dot{V}O_{2\max}$, for example, a woman's cardiac output, stroke volume, and oxygen consumption are generally less than a man's, and her heart rate is slightly higher. These relationships between HR, SV, and $\dot{Q}$ for the same absolute power output (50 W) and the same relative power output (60% $\dot{V}O_{2\max}$) are illustrated in [figure 19.3](#). These data were derived from the HERITAGE Family Study.²⁷

While several early studies reported $\dot{Q}$ to be higher in women than in men at identical submaximal power outputs, possibly compensating for their lower hemoglobin concentrations, more recent studies have consistently shown no differences between sexes.²⁷ Apparently, women can compensate for their lower hemoglobin concentrations with a steeper increase in their arterial–mixed venous oxygen difference, or $(a-\bar{v})O_2$ difference, for a given power output. Women may also have less potential for increasing their peak $(a-\bar{v})O_2$ difference with training due to their lower hemoglobin content, which results in lower arterial oxygen content and reduced muscle oxidative potential. Lower hemoglobin content is an important contributor to **sex-specific differences** in $\dot{V}O_{2\max}$ because less oxygen is delivered to the active muscle for a given volume of blood.

Sex differences in respiratory responses to exercise are largely attributable to body size differences ([figure 19.4](#)). Breathing frequency during exercise at the same relative power output (e.g., 60% $\dot{V}O_{2\max}$) differs little between the sexes. However, at the same absolute power output, women tend to breathe more rapidly than men, probably because when men and women are working at the same absolute power output, women are working at a higher percentage of their $\dot{V}O_{2\max}$. Tidal volume and ventilatory volume are generally lower in women at the same relative and absolute power

outputs, up to and including maximal levels. Again, these differences are related to their smaller body size.

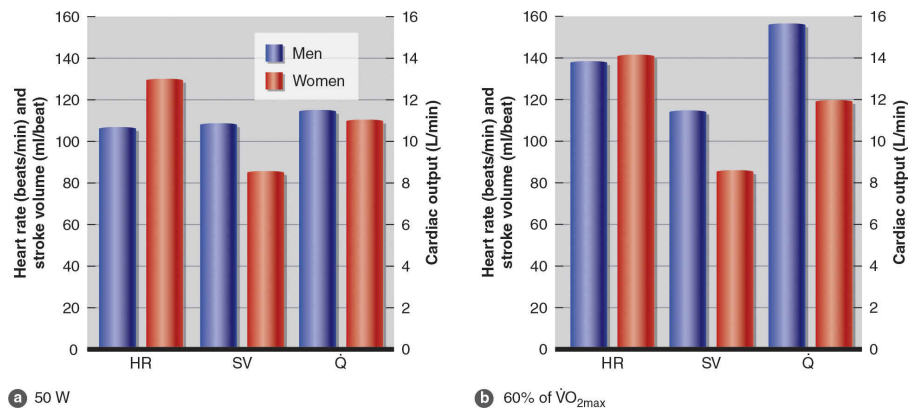


FIGURE 19.3 Comparison of submaximal heart rate (HR), stroke volume (SV), and cardiac output ($\dot{Q}$) between men and women at (a) the same absolute power output (50 W) and (b) the same relative power output (60% $\dot{V}O_{2max}$).

Based on Wilmore et al. (2001).

Maximal Oxygen Consumption

$\dot{V}O_{2max}$ is regarded by most exercise scientists as the single best index of a person's cardiorespiratory endurance capacity. The average woman's $\dot{V}O_{2max}$ is only 70% to 75% of the value of the average man's, but sex differences in $\dot{V}O_{2max}$ must be interpreted with caution. A classic study published in 1965, involving both elite athletes and nonathletes, showed that while mean $\dot{V}O_{2max}$ was lower in women, there is considerable variability and overlap in $\dot{V}O_{2max}$ within and between sexes.¹³ For example, when the subjects' physiological responses to submaximal and maximal exercise were compared, 76% of the female nonathletes overlapped 47% of the male nonathletes, and 22% of the female athletes overlapped 7% of the male athletes, as illustrated in [figure 19.5](#).

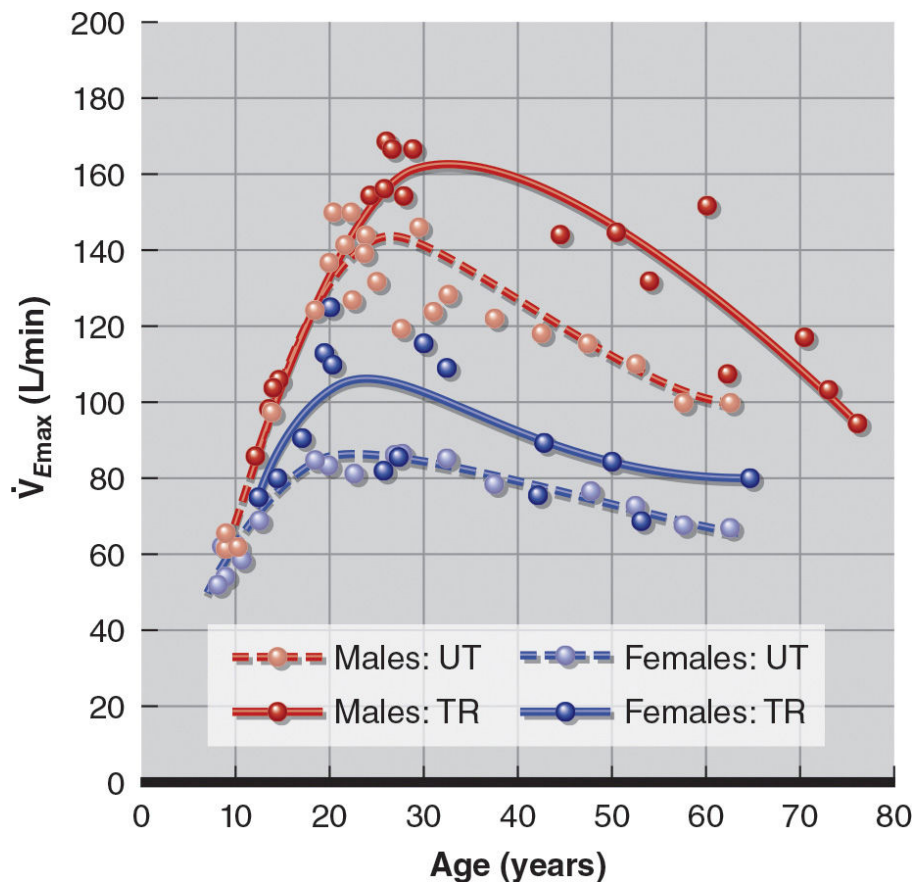


FIGURE 19.4 Differences in maximal ventilatory volumes with age in untrained (UT) and trained (TR) women and men.

Comparisons of $\dot{V}O_{2\max}$ values of normal nonathletic women and men beyond puberty are confounded by differences in the level of conditioning (i.e., nonathlete men are typically more well trained than women at the same age) as well as possible sex-specific differences. Saltin and Åstrand²¹ sought to minimize the effect of differential physical activity between sexes by comparing $\dot{V}O_{2\max}$ values of highly trained female and male athletes from Swedish national teams. In comparable events, the women had 15% to 30% lower $\dot{V}O_{2\max}$ values. More recent data, however, suggest a smaller difference of 8% to 12% between sexes, as shown in [figure 19.6](#), where $\dot{V}O_{2\max}$ values for a group of elite female distance runners are compared with values for elite male distance runners and average, nonathletic women and men.

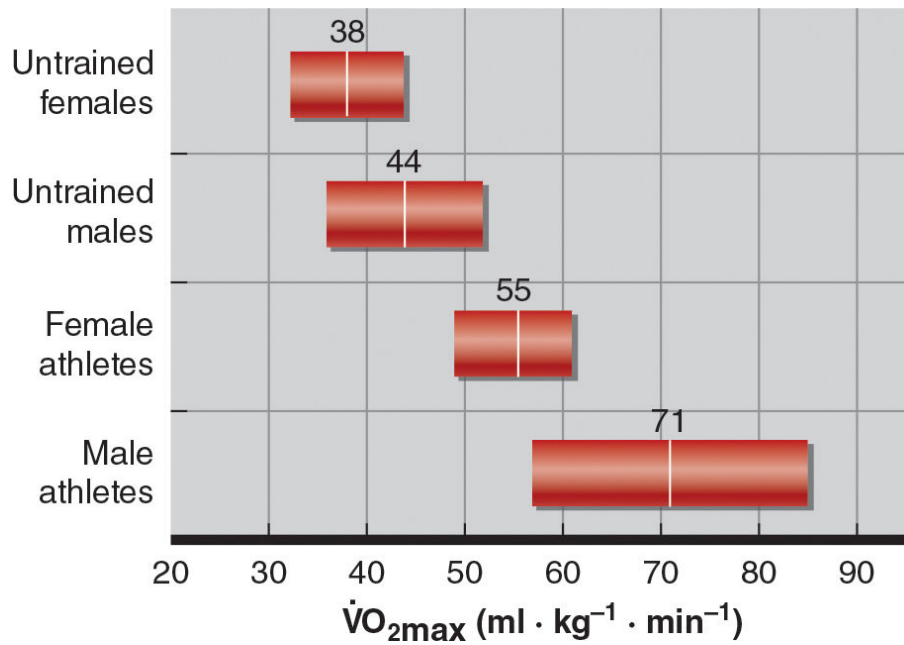


FIGURE 19.5 Range of $\dot{V}O_{2\max}$ values (mean $\pm$ 2 standard deviations from the mean) for female nonathletes, male nonathletes, elite female athletes, and elite male athletes. The mean $\dot{V}O_{2\max}$ value is presented above each bar. This figure illustrates that although differences in the average $\dot{V}O_{2\max}$ between groups can be substantial, there can be considerable overlap of one group with another. Data from Hermansen and Andersen (1965).

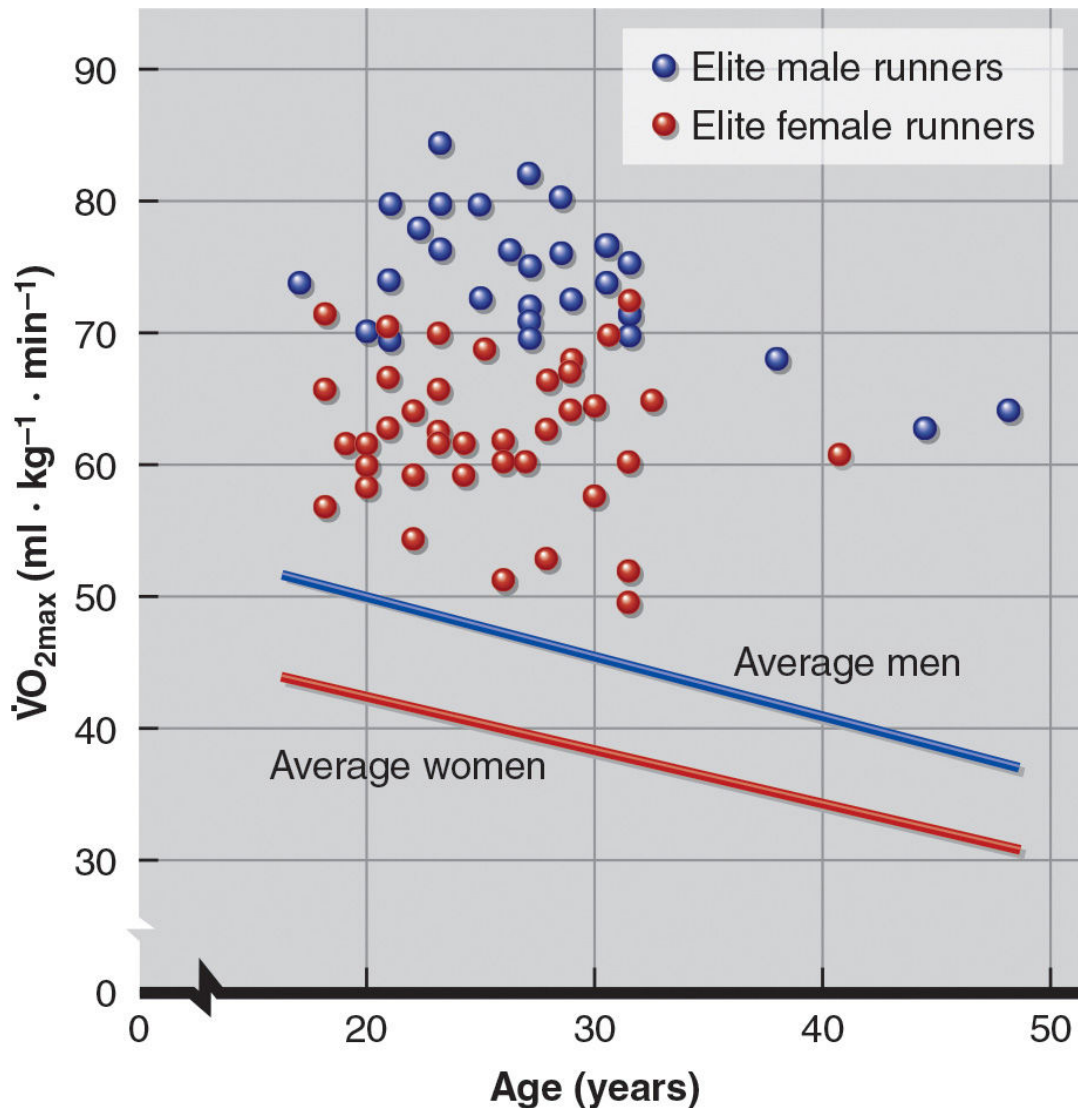


FIGURE 19.6 $\dot{V}O_{2\max}$ values, compiled from the literature, for elite female and male distance runners compared with average values in untrained women and men.

Differences between the sexes may disappear when $\dot{V}O_{2\max}$ is expressed relative to FFM or active muscle mass. For example, manipulating body weight in trained men (by adding external weight to the torso) to match the greater fat weight being carried by similarly trained women reduced the mean sex differences in submaximal $\dot{V}O_2$ (expressed in milliliters per kilogram of FFM) and $\dot{V}O_{2\max}$ by 38% and 65%, respectively.⁹ These results suggest that women's greater sex-specific essential body fat stores are major determinants of sex-specific differences in $\dot{V}O_{2\max}$ during running.

RESEARCH PERSPECTIVE 19.1

Do Men Lose More Weight Than Women with Regular Exercise?

Exercise is recommended by multiple public health organizations and professional societies as a primary method for weight control and weight loss, with the overall goal of improving health. Exercise improves health outcomes independent of weight loss, and the efficacy of an exercise intervention to produce weight loss without an accompanying diet plan is still in question (discussed in Research Perspective 15.1); however, there has been a prevailing assumption that women do not lose as much weight as men do in response to exercise. Studies that examine sex differences in weight loss often explain a lower weight loss in women by theorizing about a stronger defense of body fat resulting in greater energy intake. However, common methodological issues, including compliance to the exercise regimen, lack of actual energy expenditure data, differences in the duration and intensity of the exercise programs, and variability in subject characteristics make these studies difficult to interpret and compare.

In 2014, a research team of scientists from the University of Leeds in the United Kingdom and the Karolinska Research Institute in Sweden conducted a systematic review of studies that had examined weight loss differences between men and women.⁶ This review included studies involving short-, medium-, and long-term interventions; supervised versus unsupervised programs; and normal weight versus overweight individuals. The review also examined exercise-induced energy expenditure and compensatory increases in energy intake. Their meta-analysis indicated that when energy expenditure was equal, weight loss was the same between men and women. Additionally, the variability in weight loss responses to exercise was driven by individual variability in energy intake, independent of sex. Furthermore, there was no evidence to suggest that women compensated for exercise-induced energy-expenditure by increasing caloric intake any more than men did. These researchers speculate that previous studies showing a sex difference in exercise-induced weight loss were confounded by inattention to such variables as body size and energy intake, skewing their results.

Women's lower hemoglobin concentrations have also been proposed as a factor contributing to their lower $\dot{V}O_{2\max}$ values. However, removing a small amount of blood from men to equalize their hemoglobin concentrations to those of women reduces men's $\dot{V}O_{2\max}$ values, but accounts for only a relatively small portion of the sex difference.⁸

It is also important to understand that a woman's lower maximal cardiac output is a limitation to achieving a high $\dot{V}O_{2\max}$ value. A woman's smaller heart size and lower plasma volume greatly limit her maximal stroke volume. When we consider submaximal oxygen consumption ($\dot{V}O_2$), little if any difference is found between women and men for the same absolute power output. But since women usually are working at a higher percentage of their $\dot{V}O_{2\max}$ at the same absolute submaximal work rate, blood lactate concentrations are higher, and the lactate threshold occurs at a lower absolute power output. Lactate threshold is similar between men and women when expressed as $\% \dot{V}O_{2\max}$.

In Review

- The innate properties of muscle and the mechanisms of motor control are similar for women and men.
- Women and men do not differ in lower body strength expressed relative to body weight or to FFM. But women have less upper body strength expressed relative to body weight or FFM than men, largely because more of women's muscle mass is below the waist and women use their lower body muscles more than their upper body muscles.
- At submaximal exercise intensities, women have higher heart rates than men, but women's submaximal cardiac outputs are similar for the same intensity. This indicates that women have lower stroke volumes, primarily because they have smaller hearts and less blood volume.
- Women also have a lesser capacity to increase $(a-\bar{V})O_2$ difference, probably because of lower hemoglobin content, so less oxygen is delivered to their active muscles per unit of blood.
- Sex differences in respiratory responses are primarily attributable to body size differences.
- Beyond puberty, the average woman's $\dot{V}O_{2\max}$ is only 70% to 75% that of the average man, although these differences may be attributable to discrepant physical activity levels between men and women. Research with highly trained athletes reveals an 8% to 15% difference, and much of this difference is attributable to women's greater fat mass, lower hemoglobin levels, and lower maximal cardiac output.

- At a given relative exercise intensity, little or no difference in lactate threshold is found between the sexes.
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Physiological Adaptations to Exercise Training

Basic physiological function, both at rest and during exercise, changes substantially with exercise training. In this section, we investigate how women adapt to regular exercise, emphasizing areas in which their responses might differ from that of men.

Body Composition

With either cardiorespiratory endurance training or resistance training, both women and men experience

- a decrease in total body mass,
- a decreased fat mass,
- a decreased relative (%) fat, and
- increased FFM.

The magnitude of the change in body composition appears to be related more to the total energy expenditure associated with the training activities than to the participant's sex. More FFM is gained in response to strength training than with endurance training, and the magnitude of these gains is similar between sexes.



Bone and connective tissue undergo alterations with training. In adults, weight-bearing exercise is critical for maintaining bone mass and bone mineral density. This adaptation appears to be independent of sex. Connective tissue appears to be strengthened with endurance training, and sex-specific differences in this response have not been identified.

Strength

Until the early 1970s, prescribing strength training programs for girls and women was considered inappropriate. Women were not believed capable of gaining strength because of their innately low levels of male anabolic hormones. During the 1960s and 1970s,

however, it became evident that many of the United States' best female athletes were not doing well in international competition, largely because they were weaker than their competitors. Slowly, research began to demonstrate that women can benefit considerably from strength training programs even though the resulting strength gains are usually not accompanied by large increases in muscle bulk.

In part because of their lower testosterone, women have less total muscle mass than men. If muscle mass is the major determinant of strength, then women have a distinct disadvantage. But if neural factors are as important as or more important than size, women's potential for absolute strength gains is considerable. Also, some women can attain significant muscle hypertrophy. This has been demonstrated in female bodybuilders who have remained free of anabolic steroids. A number of studies have shown similar increases for men and women in FFM and muscle volume, as well as hypertrophy of type I, type IIa, and type IIx muscle fibers following periods of resistance training. Accordingly, women can experience major increases in strength (20% -40%) as a result of resistance training, and the magnitude of these changes is similar to that seen in men.

Cardiovascular and Respiratory Function

Major cardiovascular and respiratory adaptations result from cardiorespiratory endurance training, as discussed in [chapter 11](#), and these adaptations do not appear to be sex specific. Increases in maximal cardiac output ($\dot{Q}_{\max}$) accompany training despite the fact that maximal heart rate usually does not change (or decreases slightly), so the increase in $\dot{Q}_{\max}$ is the result of a large increase in maximal stroke volume. This results from two factors. End-diastolic volume (the amount of blood in the ventricles before contraction) increases with training because blood volume increases. In addition, end-systolic volume (the amount of blood remaining in the ventricles after contraction) is reduced with training because the stronger myocardium and increased sympathetic nerve activity produces a more forceful contraction, ejecting more blood.

At submaximal work rates, cardiac output shows little or no change with training, although stroke volume is considerably higher for the same absolute rate of work. Consequently, heart rate for any given rate of work is reduced after training. These changes are independent of sex.

Maximal Oxygen Consumption

With cardiorespiratory endurance training, women experience the same percent increase (15% to 25% on average) in $\dot{V}O_{2\max}$ that has been observed in men. The magnitude of change depends on the intensity and duration of the training sessions, the frequency of training, and the length of the study. There are also similar changes in lactate production and accumulation: Blood lactate concentrations are reduced for given absolute submaximal rates of work, peak lactate concentrations generally are increased, and the lactate threshold increases with training.

In Review

- With training, women and men experience similar changes in body composition as determined by total energy expenditure during training.
- Women, similar to men, gain considerable strength through resistance training, an effect associated with increases in FFM and muscle volume as well as hypertrophy of type I, type IIa, and type IIx muscle fibers.
- Cardiovascular and respiratory changes that accompany cardiorespiratory endurance training do not appear to be sex specific.
- Women experience the same relative increases in $\dot{V}O_{2\max}$ as men with cardiorespiratory endurance training.

Sport Performance

Women are outperformed by men in all athletic activities in which performance can be precisely and objectively measured by distance or time. The difference is most pronounced in activities such as the shot put, where high levels of upper body strength are crucial to successful performance. It would appear that the gap between the sexes narrowed as more women began to participate in sport. In the 400 m freestyle swim, however, the winning time for women in the

1924 Olympic Games was 19% slower than that for men, but this difference decreased to only 7% in the 1984 Olympics. That gap was narrowed even further by Katie Ledecky, whose 2016 Olympic record time of 3:56.46 is less than 3% away from the men's record of 3:40.14 set in 2012 by Sun Yang. As we see from [table 19.1](#), the difference between the men's and women's world records for the 1,500 m freestyle is only 5.5%. Differences are larger for track-and-field events. Unfortunately, making valid comparisons through the years has been difficult because the degree to which sport has been emphasized and women's access to coaching, facilities, and advanced training techniques have differed considerably over the years.

As noted at the beginning of this chapter, large numbers of girls and women did not start entering competitive sport until the 1970s. Once girls and women started training as hard as boys and men, their performance improved dramatically. This is illustrated in [figure 19.7](#), which shows world track records from 1960 to 2016 for women and men in six running events in track and field. For distances of 100 m through the marathon, women's present world records are consistently 8% to 9% slower than men's. Furthermore, as can be seen in this figure, the improvement in women's records, which was initially quite dramatic, is beginning to level off and to parallel the curves for men's records.

It is firmly established that there are sex differences in anaerobic and aerobic performance attributable to sex differences in muscle mass and strength, maximal oxygen uptake, and metabolism. This results in men outperforming women at every distance and event. However, trends in the performance gap between men and women have changed nonuniformly over the past 50 years. For example, whereas sex differences in ultra-endurance performance have been decreasing, the gap between male and female performance has actually increased in the sprint events. These observations suggest that factors that mediate the sex difference in performance are not solely physiological.

In the endurance events, by contrast, it may be a participation issue. For example, the gap between the top female and male 24 h ultramarathoners has decreased over the last 35 years, indicating

that as increasing numbers of women enter ultra-endurance sports, performance gaps may shrink.¹⁹ Moreover, marathon data indicate that sex differences in performance widen with age and increasing place of finisher (1st through 10th), and these relationships are mediated predominantly by lower numbers of women in competition.¹⁴

In any case, evidence to date indicates that differences between men and women in sport performance have physiological, social, environmental, and psychological influences; thus, predicting the performance gap over future generations may be more complex than simply generating a regression equation based on past performance.

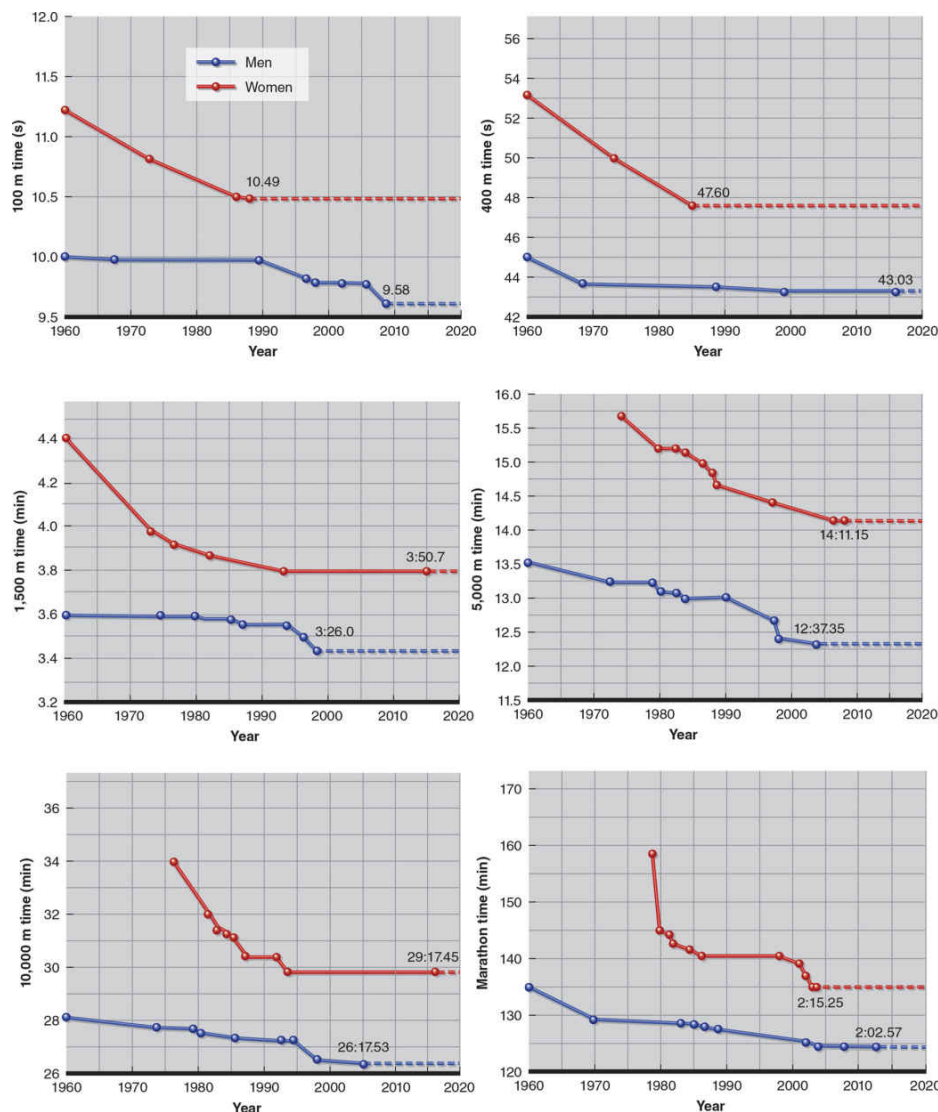


FIGURE 19.7 Women's and men's world records in six running events between 1960 and 2016.

RESEARCH PERSPECTIVE 19.2

Men Are More Likely Than Women to Slow Down During a Marathon

Optimal pacing strategies in endurance races has been a topic of interest for coaches, athletes, and exercise physiologists for some time. Traditionally, these studies have been limited to elite athletes. However, recently available data from mass-participation events have allowed investigators to study marathon running across a broader range of athletes, including nonelite distance runners. Based on these newly available data, new pacing patterns have emerged, including a finding that men are more likely than women to slow their pace as the marathon proceeds.

A 2015 study investigated these apparent sex differences in pacing in 91,921 runners across 14 marathons all around the United States.¹⁰ Researchers obtained data on sex, age, race completion time, and time to the halfway point, along with measures of racing experience (total previous races, previous marathons, fastest previous marathon, fastest 5K, and earliest year with a recorded race) when available. Pace maintenance was calculated for each participant as the percentage change in the pace observed in the second half of the marathon relative to the first half. The research team made a 12% adjustment to women's performances to account for the fact that women, on average, are 10% to 12% slower compared to similarly conditioned men.

Across the entire study, men slowed their pace 15.6% while the women slowed their pace by an average of only 11.7%. This sex difference was significant for the entire subject group across marathons, for each individual marathon, and was not influenced by finishing times (indicating that overall speed did not influence this sex difference). Slower finishing times were associated with greater slowing, especially in men. Finally, although more experience was associated with less slowing for both sexes, controlling for experience did not eliminate the sex differences in pacing.

The reason for this apparent sex difference in pacing is unclear. It may reflect physiological differences (such as a greater susceptibility to glycogen depletion and muscle fatigue in men) or simply differences in decision-making. Men, for example, may be more likely to adopt a risky pace early, i.e., they begin the race with a fast pace that they are not able to maintain across the entire course.

Special Issues

Although the sexes respond to acute exercise and adapt to chronic exercise in much the same manner, several additional areas that are unique to women must be considered. Specifically, we look at

- menstruation and menstrual dysfunction,
- pregnancy,
- bone health across the life span,
- eating disorders,
- relative energy deficiency and the female athlete triad, and
- menopause.

Menstruation and Menstrual Dysfunction

How does the menstrual cycle influence exercise capacity and performance? How do physical activity and competition influence the menstrual cycle? These are two questions of interest to exercising women, particularly female athletes.

The three major phases of the **menstrual cycle** are illustrated in [figure 19.8](#). The first is the menstrual (flow) phase, or **menses**, which lasts 3 to 5 days, during which time the uterine lining (endometrium) is shed and menstrual flow, or bleeding, occurs. The second is the proliferative phase, which prepares the uterus for fertilization and lasts about 10 days. During this phase, the endometrium begins to thicken, and some of the ovarian follicles that house the ova mature. These follicles secrete estrogen. The proliferative phase ends when a mature follicle ruptures, releasing its ovum (ovulation). The menstrual and proliferative phases correspond to the follicular phase of the ovarian cycle.

The third and final phase of the menstrual cycle is the secretory phase, which corresponds to the luteal phase of the ovarian cycle. This phase lasts 10 to 14 days, during which the endometrium continues to thicken, its blood and nutrient supply increases, and the uterus prepares itself for pregnancy. During this time, the remnants of the ruptured follicle (now termed a corpus luteum, hence the term *luteal phase*) secretes progesterone, and estrogen secretion also continues. The complete menstrual cycle averages 28 days; however, there is considerable variation in the cycle length among healthy women, from 23 to 36 days.

Menstruation and Performance

Alterations in athletic performance experienced during different phases of the menstrual cycle are subject to considerable individual

variability. There are no reliable data that demonstrate any significant physiological differences across phases of the cycle or changes in athletic performance at any time during the menstrual cycle. Elite-level performances have been achieved by female athletes during all phases of the menstrual cycle. Therefore, data gathered in the laboratory and during athletic competition concur that neither the physiological responses nor the performance of most women is substantially affected by the menstrual cycle, although women may be affected by factors such as premenstrual syndrome, bloating, and cramping.

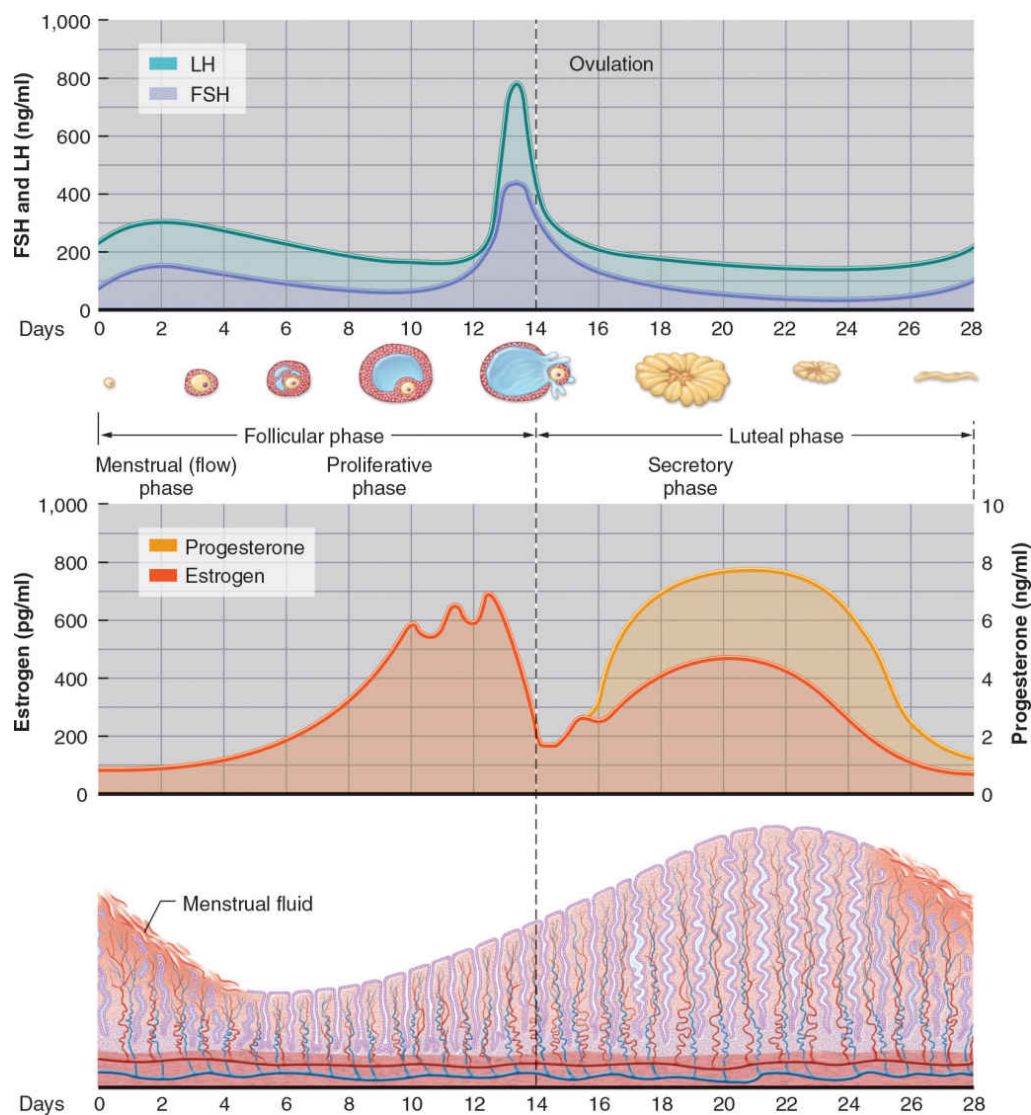


FIGURE 19.8 Phases of the menstrual cycle and the concomitant changes in progesterone and estrogen (middle) and follicle-stimulating hormone (FSH) and luteinizing hormone (LH) (top). For most

purposes, the cycle is divided into the follicular phase, which starts with the initiation of bleeding during menstrual flow, and the luteal phase, which starts with ovulation.

Menarche

Menarche, which refers to the occurrence of the first menstrual period, has been reported to be delayed in some young athletes involved in certain sports and activities, such as gymnastics and ballet. A delay in menarche is defined as a menarche after the age of 14; the median age of menarche for American girls is between 12.4 to 13.0 years, depending on the population sampled. For gymnasts, the median age appears to be closer to 14.5 years. These data imply that exercise training causes delayed menarche. However, the alternative hypothesis is that late-maturing girls, such as those with a later age of menarche, are more likely to be successful in sports such as gymnastics because of their small, lean bodies. This implies that those who naturally experience a later menarcheal age have an advantage in, and thus are likely to be involved in, certain sports rather than that their sport involvement delays menarche. At this time, evidence is insufficient to strongly support either viewpoint.

RESEARCH PERSPECTIVE 19.3

Should Female Athletes Be Tested for Iron Deficiency and Anemia?

Iron deficiency anemia impairs athletic performance, particularly in endurance sports, due to the reduced oxygen-carrying capacity of red blood cells. Iron deficiency anemia has been reported to be higher in female (<1%-18%) than in male (<1%-7%) athletes competing in a wide variety of sports. Traditionally, the higher prevalence of iron deficiency and anemia among athletes has been attributed to inadequate iron intake (e.g., restricting caloric intake in sports that emphasize leanness) and increased iron losses (e.g., via sweating, hemolysis, or menstruation).

A recent study conducted at the University of Wisconsin Madison examined the prevalence of anemia among incoming female college athletes at an NCAA Division 1 institution.¹⁸ The aims of the study were to determine what proportion of these athletes had iron deficiency anemia and what the expenses of iron-related testing would be in this setting. Medical records for all athletes on varsity team rosters were examined over a 12-year period. Anemia was defined as hemoglobin concentration below 11.6 g/dl and iron deficiency as ferritin below 20 ng/ml. A total of 5,674 total laboratory reports

were obtained from 2,749 individuals (56% female). The prevalence of low hemoglobin in female athletes at their initial physical when joining a team was approximately 6%. Across all female athletes, 2% had iron deficiency anemia and 31% had iron deficiency without anemia. Interestingly, the prevalence of anemia was not different from the average for any individual sport.

Currently, the International Olympic Committee recommends serum iron screenings as part of routine health evaluations for elite athletes. However, screening practices in the National Collegiate Athletic Association (NCAA) are inconsistent, with institutions using different biochemical indices and cutoffs to define iron deficiency anemia. In clinical testing, hemoglobin concentration and levels of serum ferritin and serum transferrin receptor are obtained in order to diagnose iron deficiency anemia. Although routine clinical screening would facilitate earlier detection and intervention in NCAA athletes, it comes with a financial cost. The median cost of iron screening exceeded \$20,000 annually for the institution.

Menstrual Dysfunction

Female athletes can experience disruptions of their normal menstrual cycle. These disruptions are collectively referred to as **menstrual dysfunction**, of which there are several types. **Eumenorrhea** is the term for normal menstrual function, referring to consistent menstrual cycle lengths of 26 to 35 days. **Oligomenorrhea** refers to inconsistent and irregular menses at intervals longer than 36 days and up to 90 days. **Amenorrhea** refers to the absence of menstruation; **primary amenorrhea** refers to the absence of menarche in girls and women 15 years of age and older—women who have not yet begun menstruating. When athletes with previously normal menstrual function report the absence of menstruation for 90 days or longer, the condition is called secondary amenorrhea. Thus, **secondary amenorrhea** is the absence of menses for 90 days or more in girls and women who were previously menstruating. Women involved in any sport and even recreationally active women can experience amenorrhea, as it occurs independent of intensity of exercise training.

The prevalence of secondary amenorrhea and oligomenorrhea among female athletes is well documented and is estimated to be anywhere between 5% and 66% or higher, depending on the sport or activity, the level of competition, and the definition used for amenorrhea. Nonetheless, the prevalence of amenorrhea in female

athletes is considerably higher than the estimated 2% to 5% prevalence for amenorrhea and the 10% to 12% prevalence for oligomenorrhea in the general (nonathlete) population. The prevalence appears to be greater in athletes who participate in activities that emphasize a lean physique, such as ballet, gymnastics, and distance running.

Since the 1970s, scientists have been conducting experiments to determine the primary cause of secondary amenorrhea in exercising girls and women and female athletes. Some of the factors that have been proposed as potential causes are the following:

- A history of menstrual dysfunction
- The acute effects of stress
- A high volume or intensity of training
- A low body weight or percent body fat
- Hormonal alterations
- An energy deficit through inadequate nutrition, disordered eating, or both

Considerable research has been conducted on each of these proposed factors and five of the six have largely been eliminated for consideration as the primary cause. It is tempting to surmise that high-volume or high-intensity training (or a combination of the two) leads to menstrual dysfunction, but this factor is likely not involved.

Current evidence indicates that inadequate nutrition resulting in an energy deficit is the primary cause of secondary amenorrhea, that is, an inadequate intake of calories such that the body is not matching caloric intake to caloric expenditure over an extended period of time. For example, research by Dr. Anne Loucks²⁰ at Ohio University and Dr. Nancy Williams at Penn State University²⁶ has clearly demonstrated that inducing an energy deficit in eumenorrheic women results in significant alterations in hormones (estrogen, progesterone, leptin, and triiodothyronine [T_3]) associated with menstrual dysfunction, including amenorrhea. Food deprivation may trigger signals that inhibit LH secretion and menstrual function, and the greater the energy deficit, the more severe the impact on menstrual function.²⁶

Thus, exercise training, per se, is likely not directly associated with menstrual dysfunction at all, other than through its contribution to an energy deficit. Rather, an energy deficit, in either the absence or presence of exercise training, is associated with these hormonal alterations. Therefore, high-intensity or high-volume training most likely is not associated with menstrual dysfunction as long as energy intake matches or exceeds energy expenditure over days, weeks, and months.

In Review

- The consequences of competing during different phases of the menstrual cycle on performance are subject to considerable individual variability. In general, there is no research evidence that demonstrates a consistent effect of phase of the menstrual cycle on sport performance.
- Menarche can occur late in some young athletes in certain sports. However, an alternative explanation for this is that late maturers, because of lean body build, are more likely to participate successfully in these activities, not that these activities cause delayed menarche.
- Female athletes can experience menstrual dysfunction, most often in the form of secondary amenorrhea or oligomenorrhea. Current evidence implicates inadequate nutrition in combination with exercise, rather than high exercise-related energy expenditure alone, as leading to a prolonged energy deficit that causes secondary amenorrhea.
- Hormonal changes associated with an energy deficit may feedback on the hypothalamus or pituitary and disrupt the normal cycle. This, too, is associated with a prolonged energy deficit.

Pregnancy

The major concern of pregnancy for active women is whether there are potential adverse effects of exercise during **pregnancy**. Over the years, several risks of exercise during pregnancy have been postulated. More recently, these risks have become balanced by the suggested and documented benefits of maternal exercise while pregnant.

TABLE 19.2 Risks and Postulated Benefits of Exercise During Pregnancy

	Maternal health	Fetal health
Risks*	Acute hypoglycemia Chronic fatigue Musculoskeletal injury	Acute hypoxia Acute hyperthermia Acute reduction in glucose availability
Benefits	Increased energy level (aerobic fitness) Reduced cardiovascular stress Prevention of excessive weight gain Facilitation of labor Faster recovery from labor Promotion of good posture Prevention of lower back pain Prevention of gestational diabetes Improved mood state and body image	Fewer complications of a difficult labor Improved cardiovascular health ⁵ Decreased risk of chronic disease ⁵

*Risks are typically low unless the patient has certain conditions, such as preeclampsia. Visit www.acog.org for details on conditions that would contraindicate exercise.

Benefits of and Recommendations for Exercise During Pregnancy

According to a 2013 review on the health benefits of exercise in pregnancy,¹⁷ physical activity during pregnancy has beneficial impacts on gestational diabetes, hypertensive disorders, excessive gestational weight gain, birth weight, timing of delivery, and child body composition. In fact, physical inactivity and excessive weight gain have been recognized as independent risk factors for maternal obesity and related pregnancy complications.¹

Unfortunately, much of the research on exercise in pregnancy has been observational, with physical activity assessed through self-report. Therefore, specific mechanisms through which exercise affects maternal and fetal well-being have not been elucidated. In any case, exercise during pregnancy can have associated risks (see [table 19.2](#)), but the benefits far outweigh the potential risks if caution is taken in designing the exercise program. In the absence of medical complications or contraindications, physical activity in pregnancy is safe and desirable, and pregnant women should be encouraged to continue or to start a program of safe physical activity. It is, however, important that the pregnant woman coordinate her exercise program with her obstetrician so that sound medical judgment can be used to determine the most appropriate mode, frequency, duration, and intensity of activity.

Despite compelling data about the substantial health benefits to being physically active during pregnancy, only about 15% of

pregnant women meet the American College of Obstetricians and Gynecologists¹ (ACOG) recommendation of 150 min of moderate-intensity aerobic activity per week during pregnancy.¹¹ Consequently, an emerging area of research has focused on the types of interventions and approaches that can increase participation in physical activity immediately before and during pregnancy.

VIDEO 19.1 Presents Jim Pivarnik on the benefits of exercise during pregnancy.



The American College of Obstetricians and Gynecologists¹ makes the following recommendations for exercising during pregnancy:

1. Physical activity in pregnancy has minimal risks and has been shown to benefit most women, although some modification to exercise routines may be necessary because of normal anatomic and physiologic changes and fetal requirements.
2. A thorough clinical evaluation should be conducted before recommending an exercise program to ensure that a patient does not have a medical reason to avoid exercise.
3. Women with uncomplicated pregnancies should be encouraged to engage in aerobic and strength-conditioning exercises before, during, and after pregnancy.
4. Obstetrician–gynecologists and other obstetric care providers should carefully evaluate women with medical or obstetric complications before making recommendations on physical activity participation during pregnancy. Although frequently

prescribed, bed rest is only rarely indicated; in most cases, allowing ambulation should be considered.

5. Regular physical activity during pregnancy improves or maintains physical fitness, helps with weight management, reduces the risk of gestational diabetes in obese women, and enhances psychologic well-being.
6. Additional research is needed to study the effects of exercise on pregnancy-specific outcomes and to clarify the most effective behavioral counseling methods and the optimal intensity and frequency of exercise. Similar work is needed to create an improved evidence base concerning the effects of occupational physical activity on maternal–fetal health.

Most activities are safe or can be modified to be safe. However, some athletic activities should be avoided,¹ including the following:

- Contact sports
- Activities with a high risk of falling
- Scuba diving
- Sky diving
- Hot yoga or Pilates

Athletic Competition and Elite-Level Training During Pregnancy

A small but growing subset of female athletes are remaining highly physically active during pregnancy and choosing to compete during pregnancy, to return to competition very quickly after delivery, or both. For example, Serena Williams won the 2017 Australian Open while pregnant and advanced to the final match of Wimbledon in 2018 just 10 months after a complicated childbirth, and U.S. Olympic discus thrower Aretha Thurmond competed in the U.S. National Championships 2 weeks after her son was born. Relatively little research exists, though, to support whether such vigorous, intense, and sustained training regimens are as safe and beneficial during pregnancy as more established patterns of moderate physical activity.

One study in 2005 followed 41 fit women through pregnancy as they participated in either high-volume (8.5 h/week of exercise) or

moderate-volume (6 h/week of exercise) training.¹⁵ Maternal and fetal safety and adverse outcomes were not different between the two groups, and women participating in high-volume exercise improved $\dot{V}O_{2\max}$ by 9.1% from 17 weeks of gestation to 12 weeks postpartum in contrast to the moderate-intensity exercisers, who only maintained $\dot{V}O_{2\max}$. Therefore, it appears that higher-volume training can actually evoke fitness gains through pregnancy in athletes accustomed to exercise before conception. There are also case reports of pregnant endurance runners who maintained training exceeding 60 mi/week (96.5 km/week) of running without any adverse outcomes and relatively few decrements in performance. By contrast, though, a more recent study of six pregnant elite Olympic-level athletes 23 to 29 weeks pregnant revealed that endurance exercise bouts at >90% maximal heart rate evoked reductions in uterine artery blood flow and fetal heart rate, indicating a potential risk to fetal well-being at from exercise of very high intensity.²³ Since data are limited to just a handful of studies on high-level athletes who maintain competitive and rigorous training during pregnancy, additional research is needed to address with certainty the questions about safety and efficacy of sustained, high-intensity training during the gestational and postpartum periods. Competitive athletes should pay particular attention to avoiding hyperthermia, maintaining proper hydration, and sustaining adequate caloric intake to prevent weight loss, which may adversely affect fetal growth.¹

RESEARCH PERSPECTIVE 19.4

Land Versus Water Exercise in Pregnancy

Healthy lifestyle habits maintained or adopted during pregnancy improve maternal and fetal outcomes, and obstetricians recommend that their patients follow healthy nutrition and physical activity guidelines during their pregnancy. An unhealthy lifestyle during pregnancy has long-term consequences on the lifetime health of the mother and the offspring, and these consequences result in a significant public health problem. The benefits of exercise on maternal and fetal health are well established, and studies of both land-based and aquatic exercise programs have reported positive results on maternal and fetal outcomes during and immediately following pregnancy. However, few studies have compared modes of exercise with respect to efficacy, safety, and compliance in pregnant women. Promoting exercise during pregnancy

can be challenging, especially in women who are not accustomed to regular exercise programs and many have concerns about safety. Because of these concerns, studies that compare pregnancy outcomes following different exercise modalities may provide pregnant women with a choice of which modality they are most comfortable, along with clinically based evidence of safety.

Researchers from Spain, Argentina, and Canada conducted a cross-sectional analysis of three separate randomized clinical trials in healthy pregnant women. This large study aimed to compare the results from studies that used (1) land, (2) water, or (3) land and water exercise programs during pregnancy. Importantly, all three of these studies used the same study design and the same exercising workloads, increasing the validity of making comparisons across studies. A total of 578 women completed the studies. For the cross-sectional analysis, there were 311 women in the nonexercise control group (pooled from all three studies), 107 women who completed the land-based exercise, 49 who completed the water exercise, and 101 who completed the land and water exercise. Total maternal weight gain, gestational diabetes rates, pregnancy-induced hypertension, gestational age, and birth weights were compared. Overall, there were no adverse effects of any of the exercise modalities. Total maternal weight gain and the percentage of women with excessive weight gain were lower in the land exercise group compared to nonexercising women. However, the number of women who developed gestational diabetes was lower in the water and water and land groups compared to no exercise. The study authors concluded that exercise performed on land is more effective for preventing excessive weight gain, whereas programs that include aquatic exercise may be beneficial for preventing the development of gestational diabetes.

More randomized controlled trials are needed to confirm these findings and explain the physiological mechanisms behind them. However, and possibly most importantly, while all exercise modalities showed some benefit, no adverse events were reported with any of the exercise modalities. These findings suggest that pregnant women can safely choose the exercise modality or combination of modalities that they find most comfortable and that they are most likely to continue throughout their pregnancy.

More than 40% of Olympic athletes in the most recent Olympiads were women, so there is little doubt that this topic will be more thoroughly investigated as increasing numbers of female athletes seek to maintain or even expand fitness and training during pregnancy.

In Review

- During exercise, major concerns for the pregnant athlete include the possible risk of fetal hypoxia, fetal hyperthermia, reduced carbohydrate supply to the fetus, miscarriage, premature labor, low birth weight, and abnormal fetal development.
 - The benefits of a properly prescribed exercise program during pregnancy outweigh the potential risks. Such an exercise program must be coordinated with the woman's obstetrician.
-

Bone Health Across the Life Span

Bone health is a particular concern for women that is affected both by physical activity and menstrual status. As they age, women are at greater risk for reduced bone mineral content and osteoporosis than men are, particularly after menopause. In addition, because bone health is connected with menstrual status in women, it is an important consideration for athletes who experience menstrual dysfunction.

Regular physical activity can play an important role in affecting bone health across the life span. In children, for example, regular physical activity is associated with greater bone mass, size, and bone mineral density, and large-scale intervention studies have shown substantial improvements in bone mineral density without an increase in fracture risk. It is particularly important for girls to engage in weight-bearing activities, given their risk of osteoporosis after menopause. The optimal dose–response and characteristics of an exercise program for bone health have not been fully described through research. School-based physical activity protocols have been successful, although they can conflict with demands on curricular time; and recent interventions showing that jumping (5-10 min of exercise performed several times a week) can increase bone mineral density by 5% to 10% have demonstrated that there are efficient ways to incorporate bone-specific physical activity into the school day. In any case, it is clear that weight-bearing physical activity can augment bone mineral density, which is particularly important for children and adolescents since growing bone has more capacity to adapt to mechanical loading than does mature bone.

Maximizing bone density early in life can help reduce the risk of osteoporosis in older women. Osteoporosis is characterized by decreased bone mineral content, which causes increased bone

porosity (see [figure 19.9](#)). Osteopenia, as we noted in [chapter 18](#), refers to a loss of bone mass that occurs with aging. Osteoporosis is a more severe loss of bone mass with deterioration of the microarchitecture of bone, leading to skeletal fragility and increased risk of bone fracture. These changes typically begin in the early 30s. The occurrence rate for fractures associated with osteoporosis increases by two to five times after the onset of menopause. Men also experience osteoporosis but to a lesser degree early in life because of a slower rate of bone mineral loss. Much remains to be learned about the etiology of osteoporosis; however, three major contributing factors common to postmenopausal women are estrogen deficiency, inadequate calcium intake, and inadequate physical activity.

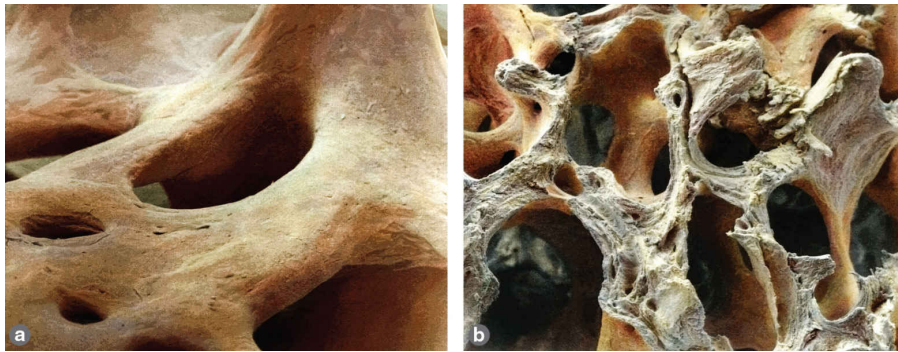


FIGURE 19.9 (a) Healthy bone and (b) bone showing increased porosity (decreased density, weakened and leached away) resulting from osteoporosis.

Although the first of these is a direct result of menopause, the last two reflect dietary and exercise patterns throughout life. Evidence suggests that in addition to the positive relationship between exercise and bone growth in children described previously, routine physical activity is associated with the maintenance of bone mass in middle-aged and older adults. For example, physically active postmenopausal women typically exhibit greater bone mineral density than sedentary age-matched women. Therefore, exercise in this age cohort protects against the normal age-related reductions in bone mineral density and thus is protective against osteoporosis and osteopenia across the life span.

In addition to postmenopausal women, women with amenorrhea and those with anorexia nervosa also suffer from low bone mass and

osteoporosis attributable to insufficient calcium intake, low serum estrogen levels, or possibly both. In studies of women with anorexia, investigators found that their bone density was reduced significantly compared with that of controls. Amenorrheic athletes also demonstrate lower bone density than normally menstruating control athletes, suggesting that physical activity does not protect amenorrheic athletes from significant bone density losses. As shown in [figure 19.10](#), the bone mineral content of normally menstruating runners tends to be higher than that of normally menstruating nonrunning controls, and female runners who are amenorrheic have higher bone mineral contents than untrained women who are amenorrheic. When we compare women of like menstrual status, those who are exercising have the higher bone mineral content.

Therefore, while evidence certainly suggests that increased physical activity and adequate calcium intake combined with adequate caloric intake are a sensible approach to preserving the integrity of bone at any age, an important point is that maintaining normal menstrual function is critical for those who have not reached menopause.

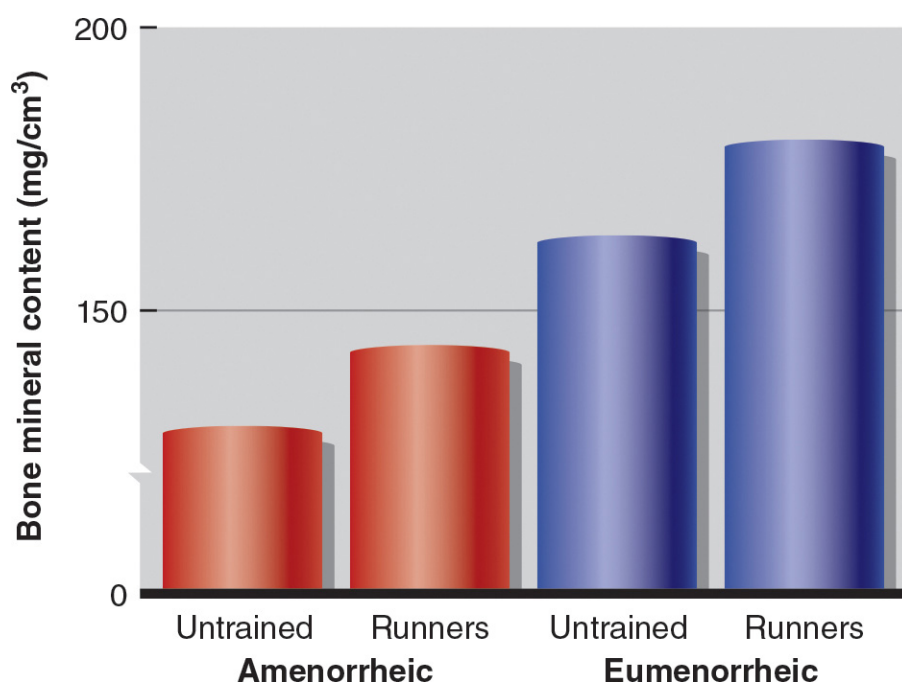


FIGURE 19.10 Bone mineral content of female runners and of untrained women who are amenorrheic and eumenorrheic. Note that when women with the same menstrual status are compared, the runners have higher bone mineral content than the untrained women.

Eating Disorders

Eating disorders are a group of disorders that must meet specific criteria established by the American Psychiatric Association. The two most commonly diagnosed eating disorders are anorexia nervosa and bulimia nervosa. **Disordered eating**, on the other hand, refers to patterns of eating that are not considered normal but don't meet the specific diagnostic criteria for a given eating disorder.

Eating disorders in girls and women became the focus of considerable attention beginning in the 1980s. Men constitute about 10% or less of the reported cases. Anorexia nervosa has been considered a clinical syndrome since the late 19th century, but bulimia nervosa was first described in 1976.

Anorexia nervosa is a disorder characterized by

- refusal to maintain more than the minimal normal weight based on age and height,
- distorted body image,
- intense fear of fatness or gaining weight, and
- amenorrhea.

Girls and women from ages 12 to 21 are at greatest risk for this disorder. Its prevalence in this group is likely less than 1%.

Bulimia nervosa, originally termed bulimarexia, is characterized by

- recurrent episodes of binge eating;
- a feeling of lack of control during these binges; and
- purging behavior, which can include self-induced vomiting, laxative use, and diuretic use.

The prevalence of bulimia in the population at greatest risk, again adolescent girls and young women, is generally considered to be about 4% and possibly closer to 1%.

It is important to realize that a person might exhibit disordered eating and yet not meet the strict diagnostic criteria for either anorexia or bulimia. As an example, the diagnosis of bulimia requires that the individual average a minimum of two binge-eating and

purging episodes a week for at least 3 months. What about the person who meets all the criteria except that bingeing and purging occur only once per week? Although this person cannot technically be diagnosed as having bulimia, her or his eating is certainly disordered and is a potential cause for concern. Thus, the term *disordered eating* has been used to refer to the behavior of those who do not meet the strict criteria for an eating disorder but who do have abnormal eating patterns.

The prevalence of eating disorders in athletes is controversial. Numerous studies have used either self-reports or at least one of two inventories developed to diagnose disordered eating: the Eating Disorders Inventory (EDI) and the Eating Attitudes Test (EAT). Results have varied because not all studies used the strict standard diagnostic criteria for either anorexia or bulimia. As in the general population, female athletes are typically at a much higher risk than male athletes, and certain sports carry higher risks than others. The high-risk sports can generally be grouped into three categories:

1. Appearance sports, such as diving, figure skating, gymnastics, bodybuilding, and ballet
2. Endurance sports, such as distance running and swimming
3. Weight-classification sports, such as horse racing (jockeys), boxing, and wrestling

Self-reports or inventories do not always provide accurate results. Secrecy is part of the behavior pattern associated with eating disorders, and people with these disorders often will not admit their problem even when anonymity is ensured. For the athlete, this need for secrecy might be heightened by fear that a coach or a parent will learn of the eating disorder and not allow the athlete to compete.

Even though research is limited, it seems appropriate to conclude that athletes are at a higher risk for eating disorders than the general population. Existing evidence likely does not reflect the seriousness of this problem in athletic populations. Although research data are not yet available, the prevalence might be as high as 60% or more in the specific high-risk athletic populations listed earlier.

The National Collegiate Athletic Association developed a list of warning signs for anorexia nervosa and bulimia nervosa ([table 19.3](#)).

When an eating disorder is suspected, it is important to recognize the seriousness of the disorder and refer the athlete to a person specifically trained in dealing with eating disorders.

Female athletes are at higher risk than nonathletes for disordered eating and eating disorders for several reasons. Perhaps most important, there is tremendous pressure on athletes, particularly female athletes, to get their weight down to very low levels, often below what is appropriate. This weight limit can be imposed by the coach, trainer, or parent or can be self-imposed by the athlete. In addition, the personality of the typical elite female athlete closely matches the profile of someone at high risk for an eating disorder (competitive, perfectionistic, and under the tight control of a parent or other significant figure such as a coach). Furthermore, the nature of the sport or activity largely dictates those at high risk. As previously mentioned, athletes in three categories are at high risk: appearance sports, endurance sports, and weight-classification sports. Added to these risks are the normal pressures imposed by the media and culture on young women, whether they are athletes or not.

Female Athlete Triad

In the early 1990s, it became apparent that there is a reasonably strong connection of three interrelated clinical disorders observed in physically active women and female athletes; the interrelatedness has been referred to as the **female athlete triad**:

- Relative energy deficiency
- Secondary amenorrhea
- Low bone mass

TABLE 19.3 Warning Signs for Anorexia Nervosa and Bulimia Nervosa

Anorexia nervosa	Bulimia nervosa
Dramatic loss in weight	A noticeable weight loss or gain
A preoccupation with food, calories, and weight	Excessive concern about weight
Wearing baggy or layered clothing	Bathroom visits after meals
Relentless, excessive exercise	Depressed mood
Mood swings	Strict dieting followed by eating binges
Avoiding food-related social activities	Increased criticism of one's body

Note. The presence of one or two of these signs does not necessarily indicate an eating disorder. Diagnosis should be made by appropriate health professionals.
Adapted from a poster distributed by the National Collegiate Athletic Association (1990).

It should be noted that disordered eating is not a necessary component of the female athlete triad. Rather, the common element is low energy availability that may or may not be a product of disordered eating. Either way, relative energy deficiency occurs as the athlete fails to consume an adequate volume of calories for her exercise energy expenditure. Over a period of time (the length of which has not been well established and might vary considerably from one athlete to another), an athlete who has low energy availability may start to experience abnormal menstrual function, which eventually may lead to secondary amenorrhea. Over time, this may lead to low bone mass. All of these conditions have been described previously within the chapter, but the female athlete triad refers specifically to the interrelatedness of the three disorders. The most recent revision of the American College of Sports Medicine position stand on the female athlete triad,² published in 2007, emphasizes that the three triad disorders can occur alone or in combination and should be addressed early before serious consequences develop.

VIDEO 19.2 Presents Nancy Williams on exercise and reproductive health and the causes of the female athlete triad.



Exercise and Menopause

Menopause is the permanent cessation of menses and is defined as occurring 12 months after the final menstrual period; it is typically reached between the ages of 45 and 55. The menopausal transition in women is driven by the declining and ultimately the loss of

production of female sex hormones, leading to cessation of menses. Of much concern is the number of menopausal symptoms experienced by women over this transition, which can be severe and disruptive to the quality of life. It has been estimated that up to 75% of women exhibit symptoms such as hot flashes, vaginal bleeding, vaginal and urinary symptoms, and mood changes. While hormone therapy (i.e., estrogen alone or estrogen + progesterone) has typically been prescribed to women to reduce menopausal symptoms, data from major clinical trials including the Women's Health Initiative indicate that hormone therapy may be tied to an increased risk of heart disease, breast cancer, stroke, and blood clots. Therefore, women with risk factors for cancer and cardiovascular disease may not wish to exacerbate risk with hormone therapy.



Exercise has been investigated as a potential alternative treatment option for alleviating menopausal symptoms, including psychological, vasomotor, somatic, and sexual symptoms. Unfortunately, there are few rigorous, well-controlled randomized clinical trials of exercise for menopausal symptoms, and cross-sectional and self-report studies have multiple methodological flaws. However, to date, it appears that exercise is effective for improving mood, insomnia, and depression in menopausal women. Its effect on vasomotor symptoms, particularly hot flashes, has not been established in large-scale studies, although in general exercise is recommended during menopause for its impacts on quality of life, body composition, and disease risk.

In Review

- Postmenopausal women, amenorrheic women, and those who have anorexia nervosa are at greater risk of osteoporosis. Physical activity and adequate calcium and caloric intake are important to the preservation of bone at any age.
 - Eating disorders, such as anorexia nervosa and bulimia nervosa, are much more common in women than in men and are especially common among athletes in appearance sports, endurance sports, and weight-classification sports. Athletes seem to be at a higher risk for eating disorders than the general population.
 - The female athlete triad comprises interrelated disorders—nutritional inadequacy, secondary amenorrhea, and loss in bone mineral density—and can be associated with negative health outcomes leading to injury and inability to compete.
 - The effect of regular exercise on menopausal symptoms such as hot flashes has not been established in large-scale studies; however, exercise is recommended during menopause because of its impact on quality of life, body composition, and disease risk.
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IN CLOSING

In this chapter, we discussed sex-specific differences in performance. Most true differences between the sexes result from women's smaller body size, lower FFM, and greater relative and absolute body fat. Making valid comparisons in sport performances has been difficult because an event's popularity and other factors—such as opportunities to participate, coaching, facilities, and training techniques—have differed considerably between the sexes over the years. Despite these limitations, we have found that female and male athletes are not as different as many people believe.

With this chapter, we conclude our examination of age and sex considerations in sport and exercise. In the next part of the book, we turn our attention from athletics to a different application of exercise physiology: the use of physical activity for health and fitness. We begin with an examination of exercise prescription.

KEY TERMS

amenorrhea
anorexia nervosa
bulimia nervosa
disordered eating
eating disorders
estrogen
eumenorrhea
female athlete triad
lipoprotein lipase
menarche
menopause
menses
menstrual cycle
menstrual dysfunction
oligomenorrhea
pregnancy
primary amenorrhea
secondary amenorrhea
sex-specific differences

STUDY QUESTIONS

1. Most physiological distinctions between men and women should be referred to as “sex differences” rather than “gender differences.” Explain.

2. How does the body composition of girls and women compare with that of boys and men? How do male and female athletes differ from male and female nonathletes?
3. What are the roles of testosterone and estrogen in the development of strength, fat-free mass, and fat mass?
4. How does women's upper body strength compare with men's? Lower body strength? Fat-free mass? Can women gain strength with resistance training?
5. What differences in $\dot{V}O_{2\max}$ exist between average women and men? Between highly trained women and men? What can explain these differences?
6. What cardiovascular differences exist between women and men with respect to submaximal exercise? Maximal exercise?
7. How does the menstrual cycle influence athletic performance?
8. What is the primary reason that some female athletes in exercise training stop menstruating for intervals of several months to several years or more?
9. What risks are associated with training during pregnancy? How can these be avoided?
10. What are the effects of amenorrhea on bone mineral? How does exercise training affect bone mineral?
11. What are the two major eating disorders, and what is the level of risk for elite female athletes having these eating disorders? How does this vary by sport?
12. What is the female athlete triad? What factors are involved, and how does the triad develop?

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.